

Layal Barakat / <https://layal.info>

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EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA

MS in Mechanical Engineering, GPA: 5.0/5.0

Graduated FEB. 2025

BS in Mechanical Engineering, minor in **Design**, ACT: 34, GPA: 4.7/5.0

Graduated FEB. 2023

Relevant Coursework: Fundamentals of Programming, AI & ML for Engineering Design, Art & Science of Negotiation, Negotiation Analysis, Management in Engineering, Developing Leadership Capabilities, Product Engineering Process

Honors and Activities: Pi Tau Sigma Honors Society (top 25% of MechE class), MIT Project Manus Makerspace Mentor

PROFESSIONAL EXPERIENCE

Graduate Researcher

SEP. 2023 - JAN. 2025

Culpepper Lab, MIT Dept. of Mechanical Engineering

Cambridge, MA

- Designed and led a novel technical makerspace training program integrating engineering design thinking, parametric CAD in OnShape, rapid prototyping, and hands-on fabrication to reduce barriers for beginner makers and underrepresented minorities.
- Conducted surveys/interviews with 23 student participants to assess changes in self-efficacy, engineering identity, and motivation.
- Utilized Pandas in Python scripts to generate histograms and box plots from Excel data to highlight pre and post training differentials.
- Awarded Best Student Paper at International Symposium for Academic Makerspaces (ISAM) 2025. Presented poster at ISAM 2024.

Mechanical Engineering Intern

JUN. 2022 - AUG. 2022

Encora Therapeutics

Remote

- Brainstormed and sketched concepts optimizing for ergonomics and patient autonomy for Encora's tremor-reducing wrist wearable.
- Developed 13 simple embedded electronic band designs in SolidWorks and analyzed stresses using finite element analysis (FEA).

Cell Engineering Intern

JUN. 2021 - AUG. 2021

Tesla, Cell Engineering Team

Palo Alto, CA

- Took ownership of existing experiment to define and understand several mechanical parameters in Tesla's 4680 tabless cells.
- Designed and executed 3 experiments to calculate battery cell pore volume over different periods of cell life.

LEADERSHIP EXPERIENCE

Design Instructor

MAR. 2025 - MAY 2025

MIT Dept. of Mechanical Engineering

Cambridge, MA

- Lectured 30 enrolled students on topics in engineering and design thinking for a first-year course in Mechanical Engineering.
- Partnered with co-instructor to revamp course content and ensure consistent course updates via Canvas.
- Organized design reviews for students with multiple stakeholders and provided feedback on student prototypes and progress.

High School Instructor

JAN. 2025

MIT Global Teaching Labs

Barcelona, Spain

- Developed Python tutorials, supported Arduino lessons, and integrated making into non-STEM course projects with up to 20 students.
- Initiated a tutorial for 4 teachers on how to use 3D printers and laser cutters for independent integration into their course curriculums.

ReVise User Research Lead

SEP. 2021 - DEC. 2021

2.009, Product Engineering Process

Cambridge, MA

- Collaborated with a 20-person team to build ReVise, a vacuum-forming vise that safely grips delicate, irregularly shaped workpieces.
- Proposed and prototyped the vacuum forming concept which defined the final product mechanism.
- Led the user research sub team to design and execute testing protocols, gather user feedback, and inform design directions.
- Presented and demoed our alpha prototype to 1000 in-person and 5000 virtual attendees.

Executive Board Member, Co-VP

JAN. 2019 - MAY 2022, JAN. 2024 - DEC. 2024

MIT Muslim Students Association

Cambridge, MA and Remote

- Established org's vision and increased collaboration with other cultural student orgs through 2024 while managing 8 committees.
- Managed a \$30k budget, directed a 10-person team across 3 org-wide committees to coordinate 30 dinners serving 100+ people daily.
- Organized 10+ events and experiences across four semesters for up to 300 people as Social Chair.
- Compiled fragmented org documentation in Google Slides to prevent information loss and streamline annual exec board transitions.

LANGUAGES

English-native, Arabic-conversational

INTERESTS

Reading, Writing, Crocheting, 3D Printing, Prosthetics, Product Design, Urban Planning, Trains, Traveling, Hiking, Pickleball, Cats